

# Derek Wang

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## EDUCATION

### University of Waterloo

Expected Apr 2030

*Bachelor of Applied Science in Computer Engineering (Co-op)*

## EXPERIENCE

### Software Engineer Intern

May 2026 – Present

*Prandtl Dynamics Inc.*

*Toronto, ON*

- Enabled real-time tracking of 5+ simultaneous moving targets at 170m range and 20ms latency by integrating SRP-PHAT acoustic beamforming with Unscented Kalman Filter + Global Nearest Neighbor multi-sensor fusion
- Cut false alarms by 40% and halved model size at 98% mAP by migrating audio detection to EfficientAT
- Eliminated cross-tenant data exposure by org-scoping queries across 6 dashboard services and WebSockets
- Hardened auth and supply chain with fail-closed guards, timing-safe key checks, and CI audit gating
- Slashed recording setup from 10 to 2 minutes by replacing a tethered rig with a Raspberry Pi service that buffers audio in memory and syncs WAV + JSON metadata into SQLite over SSH for downstream model training

### Software Engineer Intern

March 2026 – June 2026

*Kenoxis Technologies Inc.*

*Mississauga, ON*

- Lowered API latency by 35% with Redis caching across Node.js microservices on a fraud detection platform
- Designed MongoDB schemas and ingestion pipelines for job postings, improving refresh from daily to hourly
- Integrated LLM inference into backend services via async queues, reducing time-to-flag suspicious postings by 60%
- Created candidate distance and map APIs using MongoDB geospatial queries to filter applicants by location
- Accelerated onboarding of 5+ new interns by authoring internal API documentation and integration guides

### Autonomy Software Developer

September 2025 – Present

*WATonomous – Student Design Team*

*Waterloo, ON*

- Built a ROS2 navigation stack running A\* path planning over costmap grids and Pure Pursuit control under 10ms per cycle, enabling real-time autonomous rover navigation across dynamic terrain
- Led YOLO11s model retraining, improving detection mean average precision from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Optimized YOLO11s inference to 24 FPS on rover hardware via ONNX export and onnxslim optimization

## PROJECTS

### 7-DOF Robotic Arm Motion Control | C++17, Eigen, MuJoCo, CMake, React, TypeScript

- Developed 7-DOF arm kinematics and dynamics from scratch in C++17/Eigen without KDL, MoveIt, or ROS
- Achieved sub-0.2 mm tip error over 50+ scenarios via jerk-limited trajectories and 1 kHz computed-torque control
- Solved 6-DOF inverse kinematics with adaptive damped least squares and 26-seed deterministic restarts
- Prevented collisions via null-space projection of 4 secondary objectives and swept-path checking

### Autonomous Vehicle State Estimation & Tracking | C++, Eigen, ROS 2, Python, Docker, KITTI

- Implemented a 15-state error-state Kalman filter fusing 100 Hz IMU with 10 Hz GPS in C++/Eigen on ROS 2
- Localized the ego vehicle to 0.558 m mean error over 8 KITTI sequences (6.6 km), cross-checked via evo
- Wrote a 4-mode IMM multi-object tracker (CV, CA, coordinated-turn UKFs) with Hungarian data association
- Coupled tracking to localization by world-framing detections with the filter's pose (error correlation 0.976)

### Reparo - AI Repair Tool (Hack Canada 2026 Winner) | Python, FastAPI, Swift, Gemini

- Prototyped a computer-vision + Gemini reasoning pipeline producing structured repair plans from a single device photo in under 60 seconds via component detection and contextual repair inference
- Engineered a FastAPI orchestration service that normalizes unstructured LLM responses into typed repair models, enabling deterministic UI rendering of repair steps, cost estimates, and required tools

## TECHNICAL SKILLS

**Languages:** C++, Python, MATLAB, Java, JavaScript, TypeScript, SQL

**Frameworks:** React, Next.js, Node.js, Express, FastAPI, Vite

**Cloud/DevOps:** Docker, GitHub Actions, CMake

**Libraries/Tools:** PyTorch, OpenCV, Eigen, MuJoCo, ROS2, ONNX, SQLite, MongoDB, Redis, Git, Foxglove